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Research Article

Joining SiO_2 based ceramics: recent progress and perspectives

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A B S T R A C T

Nowadays, SiO_2 based material is one of the widest used materials in optical, microelectromechanical system, aerospace and some other industries. In practical application, SiO_2 based materials are required to be joined with themselves or other heterogeneous materials to assemble products with various functions. And the joining quality directly affects the mechanical performances and functional properties of assembled products. Though many researchers studied different joining technologies, explored the microstructure of joining interface and tested after-joining properties, a review that summarizes the recent cutting-edge researches and development tendency of joining technologies for SiO_2 based ceramics is still absent. Therefore, according to different applications and requirements, this review summarizes the widely used joining technologies and discusses corresponding joining mechanisms, current research progress and suitable applications. Due to the wide applications in specific industry, laser welding, anodic bonding and wafer bonding are discussed in detail. Further, brazing, the widest used method in SiO_2 joining, has been elaborated deeply in its wetting mechanism, residual stress control, interfacial microstructure and mechanical performance aspects. In the end, this review proposed the future development trends of SiO_2 ceramics joining, aiming at offering a full-view of potential improvements in SiO_2 ceramic joining technologies.

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1. Introduction

In recent years, the application of SiO_2 has greatly extended, involving in electronic, optical, medical and aerospace industry [1–3]. For example, in microelectromechanical system (MEMS), SiO_2 layers are often utilized as insulators, combining with Si semiconductors, to form Si-on-insulator (SOI) structure which largely extends the application of silicon in nonlinear optical interactions and makes it possible to process integrated circuit (IC) devices in nanoscale owing to large index contrast between Si and SiO_2 [4,5]. And in aerospace industry, SiO_2 ceramics have returned to the sight of scientists since slip-cast fused silica (SCFS) ceramics was developed which solved the inadequate mechanical properties of traditional SiO_2 ceramics. Besides, SCFS and SiO_2 matrix composites (especially SiO_{2f}/SiO_2) own the low coefficient of thermal expansion (CTE) over a wide temperature range, good wave transmissivity, thermal shock resistance, thermal stability and the lowest dielectric constant among all radome materials, which means their potential application in hypersonic speed (≥5 Ma) missile [6]. However, like other conventional ceramics, SiO_2 ceramic is hard to be directly manufactured into complex structures due to its low toughness [7]. In addition, to fulfill an integrated structure and function of products, SiO_2 ceramics need to be joined with other heterogeneous materials, such as metals or other ceramics.

Though joining of SiO_2 ceramics has been explored for decades, it still faces several severe problems and challenges. Due to its strong Si-O covalent bonding and low atomic diffusion rate, SiO_2 ceramic not only obtains its unique strengths (such as thermal stability and chemical inertness) but also by definition makes itself relatively hard to join. Even when SiO_2 ceramics are required to be joined with itself, it is hard to find a proper way to re-form interfacial atomic bonding. What makes it even more difficult is that when it comes to joining SiO_2 ceramics with metal, interfacial reactions between atoms may not take place because of different chemical bonding types [8]. Without a reliable interfacial reaction layer, it is almost impossible to achieve robust joining. Another inevitable point is that mismatch in CTE will cause large residual stress near the joining seam, especially at the SiO_2 ceramics side. Because the difference between CTE of SiO_2 (about 0.55×10^{-6}/K) [9,10] and CTE of metal (most no less than 4.6×10^{-6}/K) [11] is about an order of magnitude, the mechanical properties of samples are undermined to a great extent after the joining process. Furthermore, some devices in MEMS require that characteristics of welding or joining area maintain consistency with those of SiO_2 base

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materials, such as transparence, insulation and wave transmissivity. In complex manufacture, the consistency of materials' function needs to be taken into consideration to meet practical requirements. All stated problems show that the joining of SiO_2 ceramics requires not only enough joining strength but also the intact function as before joining. These rigid requirements make joining of SiO_2 based ceramics difficult to achieve.

In order to solve above questions, several joining technologies of SiO_2 based ceramics have been developed, such as anodic bonding [12], microwave joining [13], ultrasonic welding [14], adhesive joining [15], wafer bonding [16], brazing [17] and so on [18]. Each method has its unique advantages in a certain aspect. For example, laser welding can join transparent SiO_2 glass ceramic with itself without damage to its optical performance. In addition, SiO_2 based ceramics may contain some other components, such as Al_2O_3, BN and Na_2O [19,20], which become key factors of successful joining when interface reactions actually happen between those additives and base materials. Specifically, anodic bonding proves that dexterously utilizing the additives in SiO_2 ceramics is a viable wat to achieve successful joining. Nevertheless, the summary of these technologies which offers a guidance on joining of SiO_2 based ceramics is still absent up to now. This motivates us to write a review in SiO_2 based ceramics joining.

Based on above facts, a systematic review of SiO_2 joining is written to analyze current widely used technologies. The joining mechanisms and applications of ultrashort pulse laser welding, anodic bonding, wafer bonding and some other joining technologies are discussed. After that, brazing, which is the widest used method, is analyzed in detail through wettability, interfacial microstructure, residual stress control and mechanical performance. In the end, this review summarizes the possible future development trend according to recent researches and proposed several potential improvements in joining of SiO_2 based ceramics. This review may open some new paths in joining technology based on the full view of SiO_2 ceramics joining methods.

2. Joining technology of SiO_2

Up to now, the joining technologies of SiO_2 based ceramics have been developed rapidly. Various methods have been tried in SiO_2 based ceramics. Though there exists a certain number of research of barely used joining technologies, such as diffusion bonding, these technologies are gradually abandoned and seldom mentioned in recent years. Therefore, this chapter focuses on current popular joining technologies: laser welding, anodic bonding and wafer bonding, and ignores those obsolete methods. The joining mechanisms and current research progress of each method are analyzed in this part to show current research trend.

2.1. Ultrashort pulsed laser welding

Traditional laser welding means a linear process of photon absorption. The effect of laser is the sum of all photons. While the energy of one single photon is lower than the band gap of materials, no modification will occur. This principle allows the laser beam with suitable wavelength to pass through those transparent materials and focus on the area to be machined [21]. When radiation intensity reaches the threshold at the focus area, multiphoton process takes place as shown in Fig. 1(a), which is also called nonlinear absorption.

Based on this rule, Tamaki realized the direct joining of two transparent fused-silica glasses by ultrashort pulse (USP) laser in 2005 [22]. After that, USP laser welding of silica-based glass has been increasingly investigated. In general, USP laser can avoid many adverse effects of continuous laser. For example, USP hardly has heat affected zone (HAZ) since the ablation time is close to the time of typical electron-phonon relaxation which decides the thermal equilibrium between electrons and lattice [23]. This feature also improves spatial resolution of laser processing (smaller than 120 nm) since it removes the effect of thermal diffusion on carrier excitation [24]. In addition, contamination caused by material vapor is also eliminated because of no evaporation of surrounding materials. However, some vital problems in USP laser of silica-based glass are also exposed in the research of Tamaki [22].

First of all, interfacial gaps between joining fused-silica samples have a huge impact on welding results. Only 35 $\times$ 30 μm^2 was successful joined while 100 $\times$ 100 μm^2 was radiated in [22]. Because the melting pool is too small, it fails to fill the wide gap between the interface. To solve this problem, optical contact has been proposed as the precondition of successful USP laser welding of silica-based glass in following research [28,29]. Optical contact means that the gap between samples is small enough (can be considered as zero) to generate evanescent waves. Statistically, a typical value for optical contact is surface roughness $<$ 0.5 nm, total thickness variation $<$ 5 μm and flatness $<$ 30 μm [30]. As shown in Fig. 1(b), under the condition of optical contact, stresses that silica-based materials go through are compressive due to the expansion of melting pool in the welding process. With the solidification of the melting pool, the internal stress gradually becomes nearly zero. On the other hand, melting pool has free surface that relaxes into the gap in the heating stage but forms tensile stresses in the solidification process. Glass materials own around 20 times of compressive stress tolerance than tensile stress tolerance. As long as no cracks occur during the expansion of melting pool, optical contact gives a promise of successful USP laser welding of silica-based glass [31]. However, optical contact is one unachievable harsh condition for industrial automation production. Therefore, gap bridge way for USP silica-based glass laser welding is also needed.

Cvecek, Richter and Chen simultaneously published gap bridge methods of silica-based glass in the year of 2015 [26,27,32]. By carefully adjusted the focus position, the surface of melting pool bulged at different levels as shown in Fig 1(c). Interestingly, this volume increase is irreversible even though shrinkage occurs during cooling process, but it never neutralizes the expansion. Some scholars proved that oxygen bubbles produced in fused-silica glass contribute to this volume increase, especially when energy input is sufficiently high [33,34]. Therefore, when the bulging of interface contacted with each other, the Si-O chemical chain will intervein and form robust joining after solidification. The 3 μm gap of fused silica glass has been successfully bridged in the laboratory using a laser parameter of 10 μJ pulse burst and 0.5 ps pulse duration [27]. Among all welding parameters, laser pulse energy and frequency matter the most to bridge gaps. Increasement of each of them can join the sample with larger gaps but also raise the possibility of crack formation, as shown in Fig 1(d). Further, it has been reported that 10μm gap of soda-lime glass was joined by rapid oscillating scan [35].

Fig. 1(e) is the typical interfacial microstructure of laser welding joints. The whole welding area is a tear-drop shape. The black area in the joints is the bubble voids. With the suitable focus point, the previous gap disappeared. Observing the zoomed area in the joints, there are no any defects at the interface, demonstrating the sound joining of laser welding. Since the basic materials are amorphous, there is no the phases distribution in the whole welding interface. The interfacial microstructure is simple but is not as dense as basic materials.

In general, homogeneous glass joining by USP laser welding is one suitable technology for nano/micro-machining since it has precision up to 175 nm and extremely limited HAZ. For kinds of fused silica glass which own a low CTE, USP laser welding can achieve a joint strength (~90 MPa), which is comparable with base material

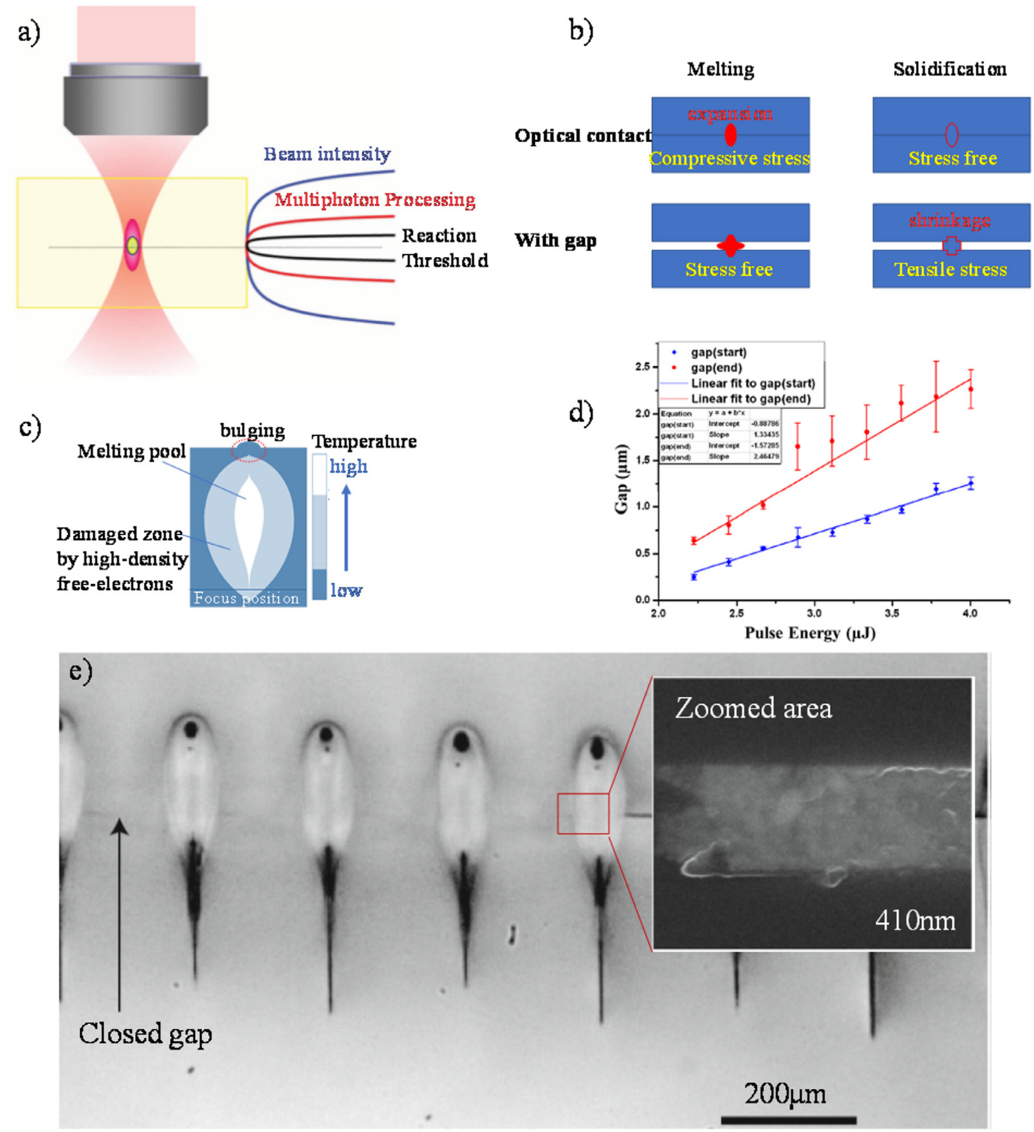

Fig. 1. (a) Theoretical mechanism of USP laser welding [25]; (b) gap influence on USP laser welding; (c) schematic illustration of USP welding area; (d) relationship between pulse energy and gap that can be bridged [26]; (e) typical interfacial microstructure of laser welding [27].

[36]. Even without optical contact, joint can reach 85% strength of pristine materials. And theoretically, the joints can withstand the temperature that basic materials can bear. However, the gap condition of less than 3 μm is still harsh to realize in industrial. Besides, the size of reported welding area still restricts on the millimeter scale. In addition, it is inspiring to see 10 μm gap has been bridged in soda-lime glass which owns a large CTE. It is explorable for oscillating scan method to apply in other silica-based glass which has a much smaller CTE than that of soda-lime glass to wider viable gap range of USP laser welding.

Besides homogeneous glass welding, USP laser welding has also been applied in joining of heterogeneous glasses. Up to now, fused silica glass has joined with SiC, sapphire and aluminum [37–39]. However, residual stress must be taken into consideration in these researches. Though successful joining has been achieved in a small size (a path of several micrometers), storage effects of residual stress in large size cannot be ignored in practical use. Therefore, there is still a long way for heterogeneous welding of USP laser welding for silica-based glasses. The complex interfacial reaction must be explained in joining of heterogeneous materials. In addition, the further improvements in gap bridging are still needed.

2.2. Anodic bonding

Anodic bonding, also known as field-assisted bonding, was proposed by Wallis in 1969 when strong bonds between metal anodes and glass under electrostatic fields was found [40]. Up to now, it has become one key joining technology of heterogeneous materials in MEMS industry. It is actually solid electrochemical diffusion reaction under the effect of temperature (usually between 350 and 450 °C) and direct current voltage (mostly from 400 V to 1000 V) to achieve permanent joining [41,42]. Specifically, it is suitable for ion-containing glass (alkali-rich content glass, such as borosilicate) heterogeneous joining [43], avoiding damage brought by high temperature in thin-film metal of precise sensor. Although anodic bonding technology can hardly apply to pure silica glass [44], other silica-based glasses which have a tremendous application in MEMS system are often chosen as base materials for anodic bonding because of alkali metal oxide additives, such as Na_2O.

Voltage and temperature are the most important parameters in anodic bonding. The alkali metal oxide starts to decompose under the effect of direct voltage and decomposed alkali cation and oxygen anion move oriented towards cathode and anode, respectively.

Table 1
Anodic bonding parameters of different materials.

Base material 1	Base material 2	Temperature(°C)	Voltage(V)	Reference
Silica-based glass	Silicon	350-400	600-1000	[45,47]
	Cu	400	1000,1400	[50,51]
	TC4	250,300	400-600	[52]
	430 stainless steel	350,400	800,1000	[53,54]

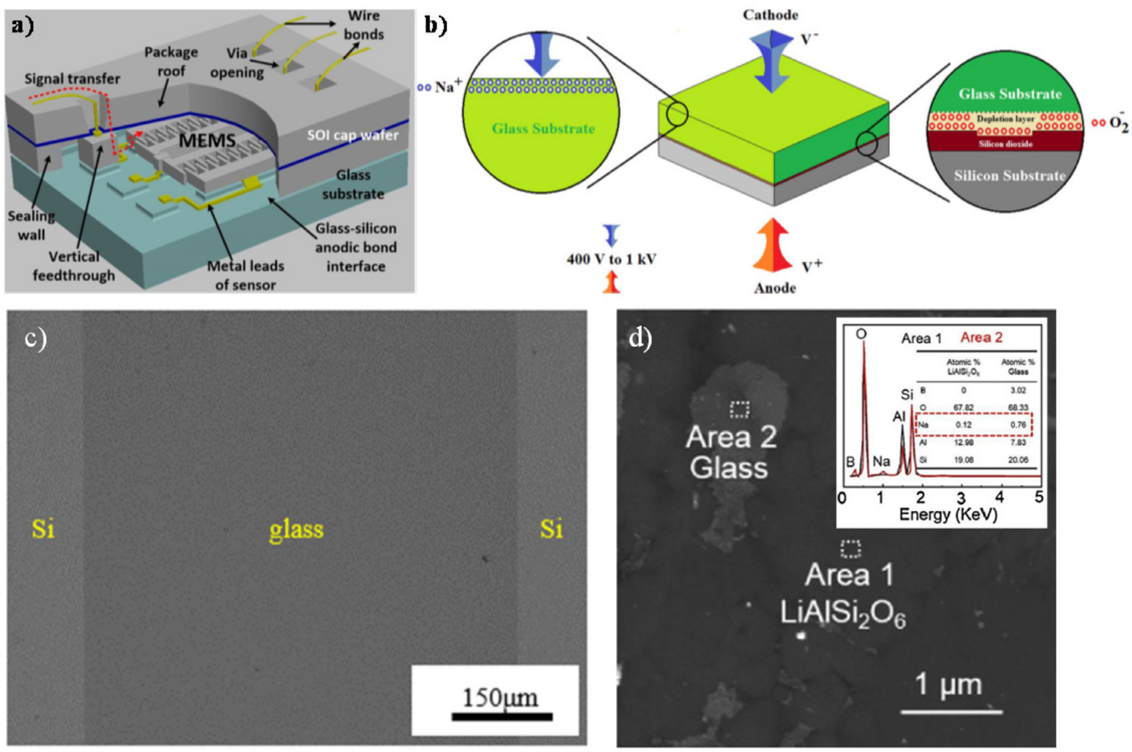

Fig. 2. (a) wafer-level application of anodic bonding in MEMS [45]; (b) scheme of anodic bonding [42]; (c) cross-section image of the anodic bonding joint [46]; (d) the digital image of back side of anodic bonding area [47].

During this stage, movement of ions is reflected at the bonding current, which can be a criterion to estimate the joining results of anode bonding [47]. This is the reason why time should not be a parameter in anodic bonding since the end of joining process is controlled by the change of currents. Temperature helps the decomposing of alkali metal oxide and makes the movement of ion easier. It reduces the threshold of voltage since a voltage higher than 1 kV will cause irreversible damage on delicate structure. In the meantime, a high temperature also induces residual of thermal stress. Generally, the joint strength increases with increasement of voltage and temperature [46]. Besides, it has been found that load can also reduce the threshold of voltage in a successful anodic bonding. But the change of load must introduce by low-efficiency pin structure. Details can be found in [48,49]. Therefore, in the application of anodic bonding, most research focuses on optimizing the voltage and temperature of anodic bonding. Some successful anodic bonding parameters have been listed in Table 1.

As shown in Fig. 2(b), a cation-depletion layer consisting of O atoms will form under the effect of voltage and temperature. This layer can be seen as a capacitor to offer electric force for O^{2-} to move to bonding interface. In addition, due to the lack of cation in depletion layer, abundant defects remain, making migration of ions much easier. This depletion layer plays a vital role in anodic bonding since all interfacial reactions occur relating to it. When silica-based glass is joined with silicon, O in depletion layer will migrate to silicon side and react with Si, forming SiO_2 amorphous layer to achieve permanent bonding. When glass is joined with metals, this layer is where interfacial reactions take place and determines joint strength. However, there is one abnormal phenomenon that alkali reaction products do occur in metal bonding. Strangely, alkali cation depletion is surely formed in glass side. There should be no alkali ions left near the bonding interface while alkali oxides are detected in the interfacial reaction products [53,54]. This result needs further explanation on the reaction paths and priority at the interface. Thermodynamic calculation and molecular dynamics simulation should be implemented to unveil the bonding mechanism. According to previous research [46,54], shear strength of joints without alkali atoms is no more than 2 MPa, but shear strength of joints with alkali atoms can reach 14.2 MPa. Therefore, further research in this aspect may largely improve the joint qualities.

Fig. 2(c) is the interfacial SEM image of the typical anodic bonding joint. Since the bonding was achieved by the reaction of depletion layer which is tiny, it is hard to directly observe the reaction products in the cross-section image. The basic materials and the glass maintain their own features, respectively. And there are no defects at the interface, showing the successful bonding. In addition, EDS linear scanning can indicate the gradient transition of the elements, which provides direct evidence of existence of depletion layer. Fig. 2(d)) is the back side image of the joint of glass-ceramic and Si. After the anodic bonding, the concentration of the alkali ions at the glass increased. This shows the migration of al-

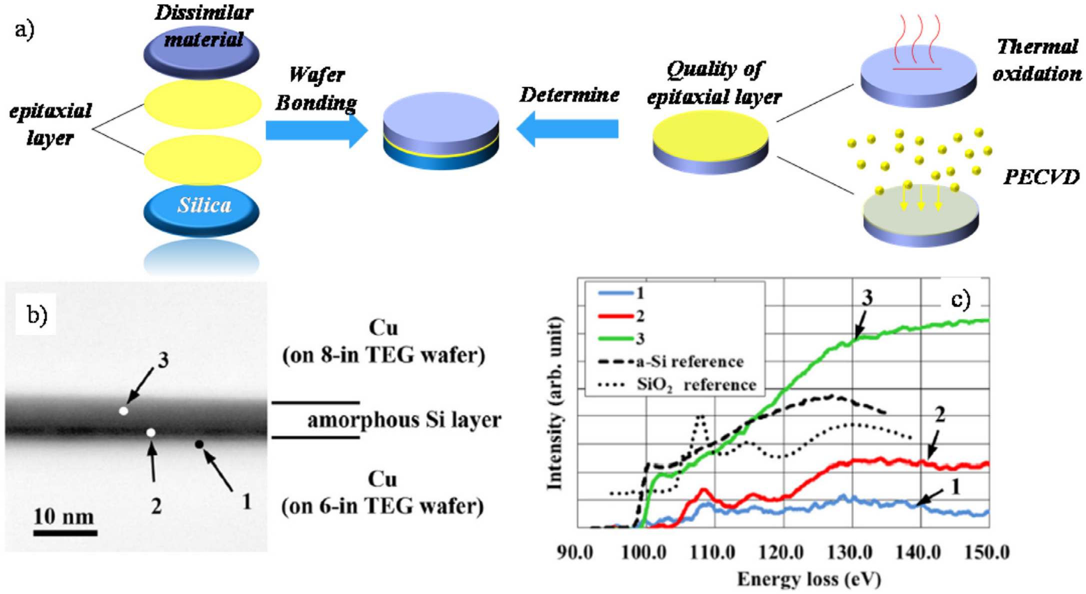

Fig. 3. (a) schematic illustration of wafer bonding; (b) TEM image of the interface of the Cu/Cu joint; (c) EELS results of (b) [55].

kali ions through the glass. Simultaneously, non-bridge oxygen and high space charge electric field are formed at the interface. Subsequently, non-bridge oxygen shifts towards silicon under the effect of electric field to bond two materials.

In short, anodic bonding aims at MEMS industry and suits for heterogenous materials joining. However, anodic bonding only suits materials that contains alkali ions. Compared with USP laser welding, it can be directly applied in wafer level sealing of MEMS device and achieve hermetic joining. Since there is no high working temperature requirement in MEMS device, the anodic bonding still limits in room-temperature application. The welding area can be up to 5 × 5 mm^2, but the biggest thickness of samples is several hundred micrometers, lower than 2 mm. Besides, anodic bonding requires a much easier surface condition than that of USP laser welding though surface state still affects the bonding results [48]. The anodic bonding technology of Si and silica-based glass is relatively mature while the joining of metals and glass still needs further improvements of microstructure control.

2.3. Wafer bonding

Wafer bonding is already a popular commercial approach for MEMS application, especially for heterogeneous material system integration [3]. Actually, the joining is achieved by polymerization reactions on epitaxial oxide layers of the mating materials as shown in Fig. 3(a). In other words, the mating material itself does not affect the results of bonding, which makes wafer bonding a universal method for joining of dissimilar materials. Strong covalent bonds between oxide layers are formed after reaction 1 and reaction 2. M stands for cation ions at the interface.

$$\text{Si-OH} + \text{OH-M} \rightarrow \text{Si-O-M} + H_2O(g) \quad (1)$$

$$\text{Si} + H_2O \rightarrow SiO_2 + H_2(g) \quad (2)$$

The polymerization reaction on the interface has been thoroughly unveiled. The joining process and thermal treatment is commercially mature. The research of wafer bonding currently concentrates on growth of epitaxial layer or expanding its application. Starting with the growth of epitaxial layer, different treatments will induce different effects. Currently, thermal oxidation or plasma enhanced chemical vapor deposition (PECVD) are two main ways of gaining SiO_2 layer. If the substrate is silicon, thermal oxidation should be the first choice since it gains a layer with smoother surface which reflects in improvement of joint strength. However, though certain surface roughness is sacrificed. PECVD approach can largely reduce treatment temperature, from 1000 °C to 300 °C, and gain SiO_2 layer on other substrates besides silicon [56]. In addition, the thickness of SiO_2 epitaxial layers needs to be carefully controlled. SiO_2 thickness was thought to be important in affecting function of samples but has a negligible effect on joint strength, but Claire proves the residual stress introduced by epitaxial growth of SiO_2 does exist and was observed by in situ curvature measurement [57].

Fig. 3(b) shows the interfacial structure of wafer bonding joints. Since wafer bonding is achieved by polymerization reactions of the surface layer, it is hard to investigate the interfacial reactions through cross-section image. Even though TEM observation was applied, no obvious reaction layer has been found. Only an amorphous layer is existed as intermediate layer. Therefore, to further prove the interfacial reactions, EDS and EELS analysis are necessary. The Fig. 3(c) shows the Si and Cu has been oxides at the interface. Through the polymerization reactions between those oxides, the bonding can be achieved. Generally, the bonding strength obtained by wafer bonding is in range of 1-5 MPa. The reported strength of Si-SiO_2 wafer bonding joints is higher than other joints, reaching ~15 MPa [55].

On the other hand, expanding of application of wafer bonding is constantly explored by researchers. To solve the large mismatch of CTE between substrates, surface-activated bonding (SAB) is largely adopted to achieve the room-temperature bonding. By introducing Ar ion beam irradiation in an ultrahigh vacuum, surfaces will become very active. The joining of surface can be directly achieved at room temperature. SiO_2 itself was proved hard to be bonded by SAB methods. However, with Fe doping, bonding of SiO_2 with SiN/$LiNbO_3$ was successful by SAB [58–60]. An extremely low concentration of Fe at the interface largely improved bonding strength by effect of recrystallization. On some other occasions where Fe is undesirable, deposition of silicon on SiO_2 surface would achieve the bonding of heterogeneous materials, such as Cu, SiC and so on [55,61,62]. The bonding quality is largely determined by deposited Si layer.

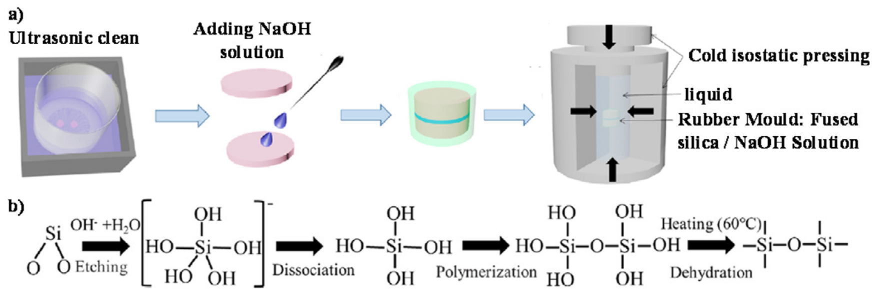

Fig. 4. (a) process of cold isostatic joining; (b) suggested mechanism for bonding formation [63].

In short, wafer bonding can achieve heterogeneous materials bonding between SiO_2 and other materials at any scale in MEMS system. Theoretically, wafer bonding has no limitation on sample size, but the sample size was restricted by equipment in practical. Now, the reported diameter of bonding sample reached 200 mm. In addition, since this method aims at MEMS system, high working temperature requirement is not applicable in wafer bonding. The reported research focuses on joints performance at room temperature. This method is more like a surface modification approach since the successful joining is foreseen as long as quality of oxidation layer is promised. Control of epitaxial layer and joining of new materials are two main research directions in this approach.

2.4. Other joining technologies

Apart from three aforementioned methods in SiO_2 based ceramic joining, some other methods were also proposed by researchers. But according to investigation of recent research tendency, these researches are either seldom studied today or brand new so that only few papers can be found. Therefore, only short descriptions are presented in this paper.

Cold isostatic joining (CIJ) is a new-developed SiO_2 joining method in 2019 by Ke [63], shown as Fig. 4. Due to the complex reactions shown in Fig. 4(b), strong Si-O-Si covalent bonds were formed at the interface, which gave its most obvious strength, insensitiveness to thermal degradation. Detailed explanation of reaction process has been elaborated in [63].

There are some points that need to mention about CIJ. First, the isostatic pressure is the key point. Without isostatic pressure, both strength and transparency of joint are largely impaired. Even when it was replaced by uniaxial pressure, the joining cannot be achieved. Over 300 MPa pressure has to be applied and samples are immersed in liquid. This is no doubt a limitation of practical application of CIJ. Second, surface roughness also affects the joining strength but does not act strongly as pressure does [64]. When surface roughness increased from 13.4 nm to 100 nm, the joining strength of samples only shows a slight drop, from 17.73 MPa to 15.2 MPa. Then, the diameter of joining samples is 15mm. For the reason of device, it is hard to realize large-scale processing and automatic production. Therefore, it is limited to potential small-size silica integration in MEMS system application.

Ultrasonic welding is another experimental way of joining silica glass and dissimilar materials. This is one solid-state joining technology which uses ultrasonic vibration to achieve permanent bonding. Welding input energy is the most important parameter in ultrasonic welding. Enough input is the precondition of joining while a large input destroys the joints. Though welding pressure and amplitude show the same effect of the joining, they cannot be adjusted as easily as energy input is. Besides, ultrasonic input energy varied over a large range in different research [65–67]. This is caused by different sizes of joining area. It is recommended that input energy density replaces the input energy as a welding parameter. However, it is worth noting that the thickness of metal side cannot exceed 1.5 mm since it influences the absorption of ultrasonic energy. And ultrasonic welding of silica-based materials is limited to spot welding which also determines its potential application. To sum up. This approach is not competitive as other approaches in silica-based materials joining.

Besides aforementioned methods, some other technologies were also tried in silica joining, such as adhesive bonding [68] and microwave bonding [13]. However, only limited research on these methods can be found since they all have corresponding insuperable intrinsic defects. For example, microwave joining needs materials as absorber of microwave to heat and melt while silica is a good wave-transmitting material. Therefore, it is hard for these attempts to be the main stream in silica joining process. More effective joining approaches of SiO_2 based materials are needed to meet the diverse requirements in the future.

2.5. Nondestructive testing of SiO_2 joints

After joining, defect detections are extremely important since they provide a direct assessment of quality of the joints. In view of time and money in evaluation and research, nondestructive tests are preferred to destructive ones. Therefore, some nondestructive technologies which are applied to evaluate joining defects are listed.

Light interference is one special nondestructive method for transparent SiO_2 materials joints. Fig. 5(a) is the schematic of light interference. When there exists unbonded areas, two reflected lights generated at the different interfaces will interfere and generate interference fringes, such as those in Fig. 5(b). This technology is extremely sensitive to small gaps. The gap of several hundred nanometers can be detected. Therefore, light interference technology is extremely helpful in detecting the unbonded area. However, this technology only suits for transparent materials.

X-ray imaging technology is another nondestructive method for defects detection. Utilizing different absorption of X-ray in the sample, clear image contrast is formed. X-ray imaging test is sensitive to volume defects but is not sensitive to interfacial problems. It is worth noting that X-ray imaging testing is now capable of 3-dimension reconstruction of defects as shown in Fig. 5(c) [70]. This breakthrough makes X-ray testing now can be applied in the joints of SiO_2 based composites. Since traditional X-ray image is 2D imaging, the defects that exist in the basic materials, especially the void in composites, will largely reduce the creditability of X-ray testing results. Therefore, this method can effectively detect the voids in the SiO_2 composites joints.

Ultrasonic evaluation is also viable in defects detection of SiO_2 joints. As shown in Fig. 5(d), by collection the signal of the ultrasonic waves reflected at the interfaces, this method can effectively distinguish different kinds of defects from the whole joining seam. However, this method needs detectors to scan all over the sample

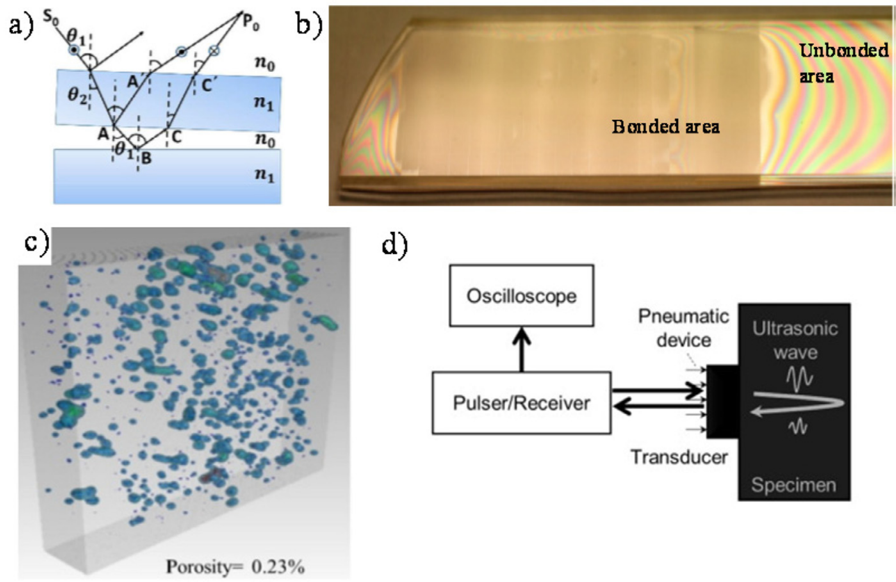

Fig. 5. (a) schematic of light interference [69]; (b) defects testing result of light interference [27]; (c) 3D-reconstruction of defects by X-ray [70]; (d) schematic of ultrasonic evaluation [71].

Table 2
Nondestructive evaluation technology of SiO_2 joints.

Methods	Type of defects that can be detected	Advantage	Disadvantage
Light interference	All	Sensitive	Basic materials need to be transparent
X-ray	Voids	Results are clear	Only viable for voids with enough sizes
Ultrasonic	All	Suitable for all materials	Hard to process

surface, and data processing is relatively complex. In general, all three technologies are summarized in Table 2.

3. Brazing of SiO_2

Among all silica joining technology, brazing is the most widely used way and is viable in different applications. By melting brazing filling materials which will achieve bonding at atom level by chemical reaction, dissolution or diffusion, brazing does not require base material to melt so that the properties of base materials can be kept at the maximum level. Therefore, even in functional materials joining, as long as suitable brazing filling materials were developed, brazing is one promising way to achieve robust joining. With the advantages of good joining strength and tailoring specific function, brazing can meet almost all SiO_2 based ceramics joining needs.

3.1. Wettability of SiO_2

The process of brazing is actually the sprawling and reaction of melting brazing filling materials. Corresponding sprawling extent of brazing filling materials on base materials is called wettability of brazing filling materials. A good wettability is the precondition of a good brazing joint. Therefore, wetting is chosen as the first part of brazing. The assessment of wetting behavior includes contact angle tests, spreading area tests and theoretical calculations. In real brazing researches, contact angle, shown in Fig. 6(a) and (b), is chosen by most researchers for its convenience and accuracy. According to Young's equation,

$$\gamma_{sg} - \gamma_{sl} = \gamma_{lg} \cos\theta \tag{3}$$

γ_{sg} is solid-gas interface tension. γ_{sl} is solid-liquid interface tension. γ_{lg} is liquid-gas interface tension and θ is the contact angle. Usually when contact angle is lower than 90°, the brazing filling materials are thought to wet base materials. The lower contact angle is, the better wettability is. However, Young's equation only suits for smooth surface. To expand the application range of Young's equation, the contact angle can be amended by Cassie-Baxter wetting model, defined as following equation:

$$\cos\theta_c = \gamma \cos\theta_y + f - 1 \tag{4}$$

γ is surface roughness factor. θ_y is Young's contact angle on smooth surface. θ_c is contact angle on real rough surface and f is rate of wetting area on whole surface. When f is getting close to 1, the equation is simplified to Wenzel equation:

$$\cos\theta_w = \gamma \cos\theta_y \tag{5}$$

γ is surface roughness factor. θ_y is Young's contact angle on smooth surface. θ_w is contact angle on real rough surface. But it needs to know that the complex reactions and interactions between base materials and brazing filling materials are ignored when assessing the wettability. Therefore, wettability experiments cannot give a precise analysis but are still enough to show brazing filling materials wetting behavior in brazing process. In general, the contact angle needs to be as low as possible to ensure the flowability of melting brazing filling materials.

Regarding the wettability on silica surface, AgCuTi brazing filling material is the first place to be discussed since it is the widest used brazing filling material in silica brazing. Active metal element Ti in AgCuTi, which can rapidly react with SiO_2, is the key to good wettability. The difference of intrinsic property between covalent bonds in silica and metal bonds in brazing filling metals makes wetting behavior hard to occur. This is proved by different wetting results of AgCu brazing filling material and AgCuTi brazing filling material on SiO_{2f}/SiO_2 surface. Fig. 6(c) shows the 135° contact angle of the AgCu on SiO_{2f}/SiO_2, demonstrating the poor wettability

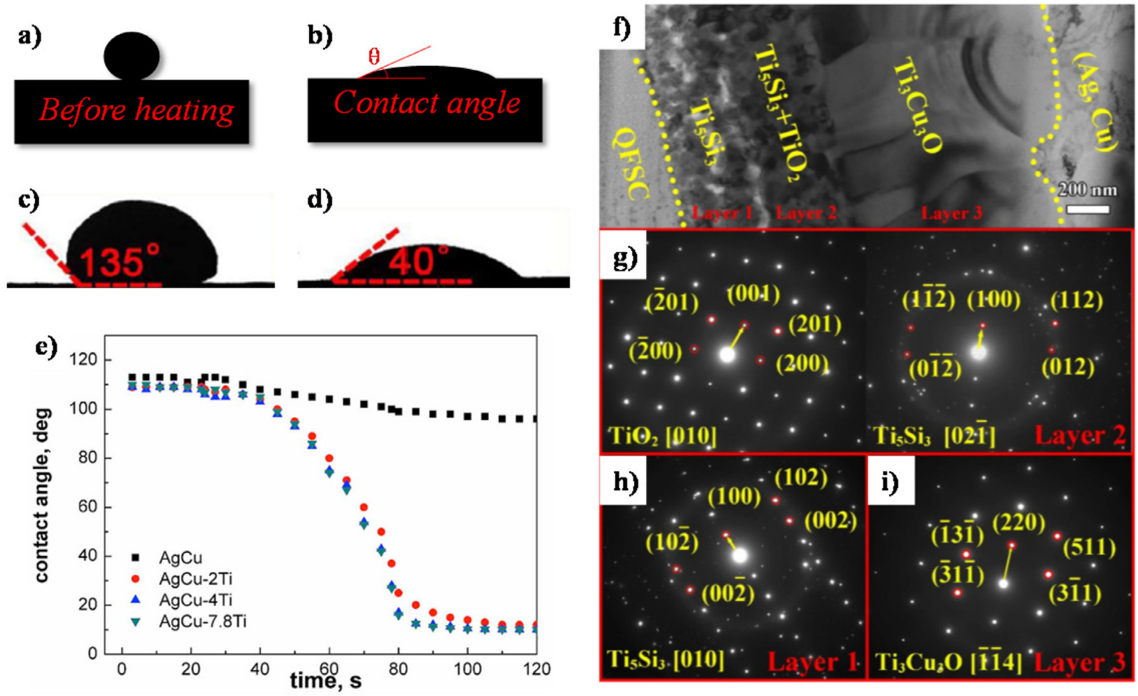

Fig. 6. (a, b) schematic illustration of contact angle; (c) AgCu contact angle, (d) AgCuTi angle [72]; (e) contact angle changing plot with different contents of Ti [73]; (f) TEM image of reaction layer, (g–h) corresponding diffraction pattern of (f) [78].

of AgCu on silica surface while AgCu+1wt%Ti exhibits a good wettability on SiO_{2f}/SiO_2 surface with the contact angle of 40° [72]. Because of the chemical dissimilarity, liquid metals do not intrinsically own a good wettability on oxide (SiO_2) surface, which further attests to necessity of active metal elements in brazing filling materials. Based on this fact, how Ti contents affect wetting behavior on SiO_2 ceramic was studied [73]. Fig. 6(e) compares the final stable contact angles at 1173K. Initial 2wt% additions of Ti into AgCu have a remarkable improvement on wettability, reducing the contact angle from 95° to 10°. However, following additions of Ti contents, from 2.0 wt% to 7.8 wt%, do not form a further drop in contact angle. To further explore the function of active metal Ti, the phases at the wetting interface were identified by TEM test. Added Ti element reacted with SiO_2 and formed Ti_5Si_3 and Ti_4Cu_2O thin layer at brazing filling material and silica side, respectively. This is the reason for improvement in wettability since wetting behavior actually took place on reaction layer instead of silica surface. According to Young's equation, both γ_{sg} and γ_{sl} are changed in wetting process leading to a totally different wetting result. However, there exists the inconsistent interfacial microstructure in [74]. The result shows an opposite situation that the reaction layer next to SiO_{2f}/SiO_2 is the combination of Ti_5Si_3 and TiO_2, and reaction layer next to brazing alloy is the mixture of Cu_3Ti_3O and Cu_4Ti_2O. This difference may be caused by the complex interface reactions. TEM can only observe one extremely small area of whole wetting interface and wetting process involves a series of complex nonequilibrium thermodynamics reactions, so it is possible that separations and movements of phases in chosen area do not finish into the final state. Since the later result accords with most of wetting research on SiO_2 [74–79], as shown in Fig. 6(f–i), it is more credible and reliable. Here, formation of $Cu_xTi_{6-x}O$ layer is believed to be the vital factor in AgCuTi's wetting behavior.

Nevertheless, sometimes it cannot gain satisfying wetting results by single addition of active metal elements. For SiO_{2f}/SiO_2 composites, the final contact angle of AgCuTi is not low enough, even on $Cu_xTi_{6-x}O$ reaction layer. Unsmooth spreading of melting metals is the reason for poor wettability in origin SiO_{2f}/SiO_2 surface. Because gaps between SiO_2 fiber and SiO_2 matrix surface of SiO_{2f}/SiO_2 cannot be as flat as pure SiO_2 ceramics surface, consecutive $Cu_xTi_{6-x}O$ reaction layer ends at these gaps. Therefore, the melting metals cannot spread smoothly. To further improve wetting behavior of brazing filling materials, surface modification was applied in SiO_{2f}/SiO_2 joining. According to previous studies [80,81], Ti has a good affinity for C element. Therefore, introducing carbon layers at the surface of composites can largely improve the initial wetting behaviors of Ti-contained brazing filling materials. In the year of 2017, coating carbon particles on ceramic surface by methane plasma treatment was observed to enhance the spreading of AgCuTi on silica [75]. XPS results show that part of Si-O bonds at surface will change into Si-C bonds, which demonstrates the successful surface modification. More importantly, due to the relatively low content of carbon along the whole brazing seam, the interfacial microstructure will not be affected by carbon while the wetting behavior can be largely improved. Similarly, phenol-formaldehyde resin was used as the carbon source to modify the surface of silica ceramics [76]. By a simple anneal process, a continuous carbon layer was formed through carbothermal reduction reaction. In addition, anneal process can also fix the prepared graphene on silica surface [77]. However, the thickness of coating layer is hard to control through annealing process. This often leads to a thick carbon layer that will affect the interfacial reactions. These three methods can effectively fill the gaps of SiO_{2f}/SiO_2 and form a carbon material layer to improve the wetting behavior. The results are summarized in Table 3.

Further, if brazing filling materials can fill the gap itself, a 3D pinning structure of brazing seam will form to largely improve joints strength. Therefore, vertically aligned carbon nanotubes (VA-CNT) modified method was proposed to improve AgCuTi wettability on SiO_{2f}/SiO_2 [78]. After a complex process of VA-CNT modification, liquid metals can permeate into gaps instead of getting hindered by those gaps because capillarity of smaller paths between VA-CNT offers additional force to make melting alloy

Table 3
surface modification of SiO_{2f}/SiO_2 (AgCuTi brazing filling material).

Approach	Coating layer	Thickness	Final contact angle	Defects
Anneal	Amorphous carbon (Or Graphene)	Thick	43° (42.8°)	Influence on interfacial reactions
Plasma treatment	Carbon particles	Thin Discontinuous	27°	Uneven thickness
PECVD	VA-CNT	Thin	30.6°	Complex processing Low efficiency

spread smoothly. After modification, final contact angle decreased from 96.5° to 30.6°. In addition, the author elaborated the reaction products in AgCuTi wetting process, shown in Fig. 6(f–i). Ti mainly concentrated on the reaction layer and formed Ti_5Si_3, TiO_2 and Ti_3Cu_3O layers which promote the spreading of brazing filling materials while Ag and Cu remain at initial position. From brazing metal to silica base material, the reaction layers are Ti_3Cu_3O, $Ti_5Si_3+TiO_2$ and Ti_5Si_3 in sequence. This further proves the low contents of carbon at the interface since no obvious carbon-related products were found. Though permeating of AgCuTi at SiO_{2f}/SiO_2 was also observed in [72], the controllability and stability of VG-CNT are much better.

It is worth noting that the crystal structure of SiO_2 has an influence on wetting behavior of AgCuTi. The reaction products on crystallized silica were Ti_3O_5 and Ti_2O_3, leading to a contact angle of 87°. On the contrary, Cu_3Ti_3O was formed near brazing filling materials side on the amorphous silica, leading to a contact angle of 43° [79]. This phenomenon explains that the difference in contact angle of AgCuTi on silica is the result of different crystal statuses. When the SiO_2 matrix and SiO_2 fiber own different crystal structures in SiO_{2f}/SiO_2 composite, surface etching oriented toward crystalline phase can improve wetting behavior of AgCuTi [82,83]. Even though this method can easily be achieved by hydrofluoric acid, its defects are also obvious. The controllability is relatively low and it only suits composites with different crystal phases. It is not as practical as other approaches in Table 3. However, if the reaction mechanism and reaction path at the interface can be unveiled according to different structures, the further optimization of wetting control should appear.

Besides AgCuTi brazing filling material, CuTi [84] and TiZrNiCu [85] alloys are also studied to wet silica surface. The wetting reaction processes show no difference with process of AgCuTi. The formation of $Cu_xTi_{6-x}O$ is also the key reaction product to improve wettability, so there is no need to further discuss the wetting situation of these two brazing filling materials in detail.

In general, active metal, Ti, are vital factors in wetting on silica. There are 4 steps in wetting process of Ti on silica. First, liquid Ti sits on the surface without chemical reaction. Then, the Ti-O and Ti-Si layer formed. After that, discontinuous $Cu_xTi_{6-x}O$ appears and spreads over the wetting area. Lastly, liquid AgCuTi wets on silica surface. Carbon materials modification is a promising way to improve wetting behavior on silica surface, especially on SiO_{2f}/SiO_2 composite. Different methods can be chosen to decide whether carbon affects interfacial reactions or not. This is extremely important in interfacial microstructure control.

3.2. Typical interfacial microstructure of SiO_2

The wetting is the macro performance of the microstructure influence. And the mechanical performances of joints are also largely determined by interfacial microstructure, such as distribution of reaction products. Therefore, it is essential to study the microstructure of joints which gives a direct guidance on how to optimize the brazing parameters and offers a clear clue for mechanism of strength improvement. In silica brazing, most current research focused on heterogeneous materials. According to different properties of base materials, this research divides the interface microstructure into following three classes as Table 4.

The first category is the material which hardly dissolves in brazing seam or has an extremely limited reaction area at the brazing filling materials interface. In this situation, the interface microstructure is mainly affected by brazing filling materials' chemical components and corresponding reactions with SiO_2. The most typical base material is Nb. Fig. 7(a) shows the interfacial microstructure of the joint of SiO_2-BN and Nb by TiNi [80]. Nb was dissolved and formed (Ti, Ni, (Nb,Ti)) ternary Layer I. The whole joint is made up of Nb/(Ti, Ni, (Nb,Ti))/TiNi/TiNi + Ti_3O_5 + Ti_5Si_3/SiO_2-BN. Fig. 7(b) demonstrates that Nb does not react with AgCuTi brazing filling materials, either. It can be clearly seen that Nb has a limited diffusion distance in brazing seam, Ag(s,s) and Cu(s,s) occupy most place of seam. In this situation, interfacial microstructure shows no difference with wetting microstructure. Besides Nb, 30Cr3 high-tensile steel and 316L stainless steel show a similar tendency in brazing with SiO_2. The only effect of 30Cr3 steel on brazing seam was that it consumed a negligible amount of Ti in AgCuTi to form $TiFe_2$ phase marked D in Fig. 7(c) [9]. The rest AgCuTi would follow same pattern as wetting experiments to form reaction layers. Similarly, when Weber studies the joining of SiO_2 and 316L steel by Bi-Ag based brazing filling materials, the interface microstructure shows no obvious relations with 316L steel [86]. In this category, the interface microstructure is determined largely by chemical components of brazing filling materials. The various contents of additives or existing elements lead to different reaction products and corresponding diverse distributions of different phases in brazing interface.

The second category refers to the material which can provide active metal elements to react with SiO_2. The Ti6Al4V (TC4) is the representative metal of this category. The direct joining of TC4 and SiO_2 by AgCuTi will be largely influenced by excessively dissolved Ti of TC4 [87,88]. Fig. 8(a) shows the interface microstructure consisting of $TiSi_2$, Ti_4O_7, TiCu, Cu_2Ti_4O, Ti(s,s), Ti_3Al and some other Ti contained phases. Different Ti contained phases permeate over the whole interface. And the same situation occurred when brazing filling materials changed from AgCuTi to TiZrNiCu or AgCuInTi. Excessive Ti element leads to too thick reaction layer which largely limits the mechanical performance of joints (maximum shear strength is around 20MPa). To solve this problem, surface modification of graphene oxide (rGO) can consume excessive Ti in brazing seam and limit brittle reaction layer's growth [77]. In Fig. 8(b), it can be clearly seen that TiC particles exist in brazing seam, proving that rGO is viable in control excessive Ti by forming TiC. Further, inserting foam metal is also effective in consuming Ti [81,89]. The effects of Cu foam and Ni foam were proved to be effective interlayers for controlling Ti contents. Due to original 3D framework of foam metal, reaction products, Ti-Cu or Ti-Ni compounds, were evenly distributed in the brazing seam, which avoided the occurrence of continuous brittle phases. Lastly, by eliminating active metal element Ti in brazing filling materials, dissolved Ti from base materials compensates for lack of active metals in brazing seam [90]. In Fig. 8(c), while Ni layer first reacted with Ti in layer A, a reasonable amount of Ti passed layer A and continued to react with SiO_2 to realize permanent joining. This method greatly increased the mechanical performance of joints from around 20 MPa to 110 MPa. Active brazing does not suit

Table 4
Category of interfacial structure according to base materials.

Category	Typical materials	Character
I	Nb, 30Cr3	Hardly involved in interfacial reaction
II	Titanium alloy	Offering excessive active metals (Ti)
III	Invar	Intensive inclination to react with active metals

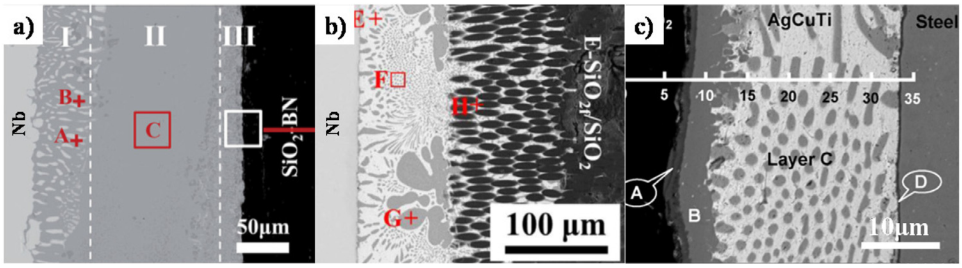

Fig. 7. joint microstructures of (a) Nb and SiO_2-BN [80]; (b) Nb and SiO_{2f}/SiO_2 [83]; (c) 30Cr3 and SiO_2 [9].

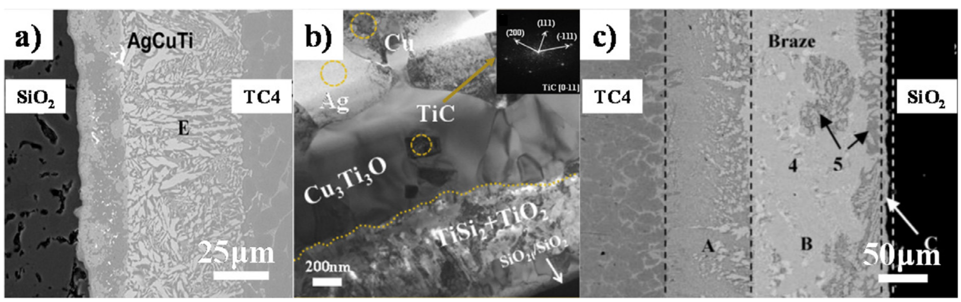

Fig. 8. typical interfacial microstructures of SiO_2 and TC4 brazing seam. (a) AgCuTi brazing filling materials [87]; (b) TEM image of joints with rGO [89]; (c) AgCu/Ni brazing filling materials [90].

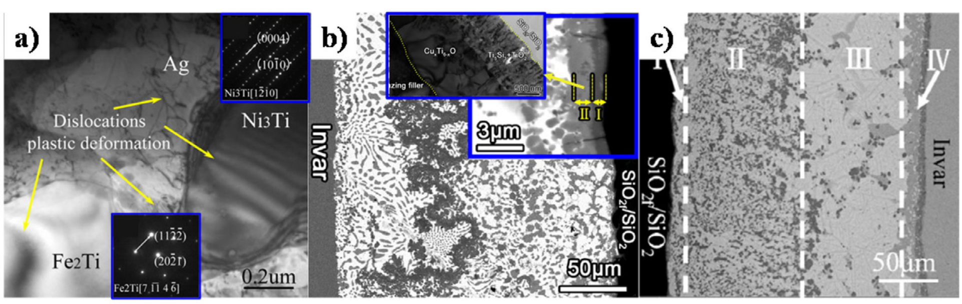

Fig. 9. (a) brittle phases in invar brazing seam [19]; (b) whole interfacial microstructure of invar brazing seam [91]; (c) graphene improved interface [84].

this kind of materials. Instead, methods that limit the Ti contents are essential to ensure a good interfacial microstructure.

The last category is the material which will largely consume active metal element Ti in the brazing filling materials. Invar alloy is the typical one in this category. Fe and Ni elements can dissolve into brazing seam and consume active metal element Ti in brazing filling materials. This conclusion was found in the direct joining of Invar and SiO_2-BN by AgCuTi [19]. As shown in Fig. 9(a), Ni_3Ti and Fe_2Ti were found in the brazing seam, demonstrating that Fe and Ni from invar alloy will consume Ti and form brittle phases. The formation of continuous brittle phases will largely impair joint strength. Further, Fig. 9(b) elaborated the microstructure of this brazing seam [91]. On SiO_{2f}/SiO_2 side, Ti_5Si_3 $+Ti_2O$ layer was formed with $Cu_xTi_{6\text{-}x}O$ layer close to it. And Fe_2Ti and Ni_3Ti compounds were linked together spontaneously and formed a chain-like shape phase in brazing seam. To inhibit the formation of brittle phase, the barrier was inserted at the interface. AgCu/Cu/AgCuTi multilayer structure was introduced to avoid formation of too many brittle Fe-Ti and Ni-Ti compounds [92]. 100 μm middle Cu layer effectively inhibited formation of brittle phases on SiO_2 side. Similarly, W interlayer can split the reaction layer into two half areas because W is highly inert and can isolate Fe and Ni elements on invar side. Therefore, inserted barrier separated the reactions and stopped the diffusion of elements at the metal side. In addition, graphene can also suppress the formation of brittle phases [84]. Due to the strong affinity between Ti and C, Ti tends to react with C at the ceramic side. Fig. 9(c) clearly shows the disappearance of brittle phases aggregation.

It is worth mentioning that the first principle calculation started to apply in SiO_2 brazing process. For example, Lu and La atoms were calculated to be stable at the Sn/SiO_2 brazing interface [93]. Though complex chemical reactions and interactions between phases pose a big obstacle on simulation, simplified model can still provide guidance on experimental chosen or mechanism explanations. It is highly recommended to use more simulations in brazing analysis to offer a clearer reaction process (such as reaction priorities and starting point of reaction) rather than single thermodynamic calculation.

In this part, according to different reaction properties of brazing filling materials, the microstructure evolution of SiO_2 based ce-

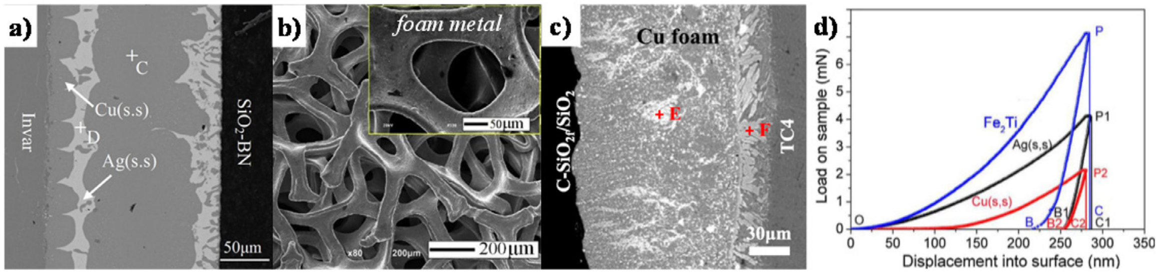

Fig. 10. (a) interfacial microstructure of Cu foil inserted brazing seam [92]; (b) framework of foam metal [94]; (c) interfacial microstructure of Cu foam inserted brazing seam [89] (d) Load-Displacement curves of different phases [72].

Table 5
Lists of residual stress control through deformation.

Base materials	Additional deformation material	Increasement of shear strength
SiO_2-BN and Invar	Cu foil	207% (compared with pristine AgCuTi)
SiO_{2f}/SiO_2 and TC4	Cu foam	200% (compared with pristine Cu foil)
SiO_2-BN and TC4	CNT reinforced Ni foam	145% (compared with pure Ni foam)

ramic brazing joints has been divided into three parts. This provides corresponding solutions for different situations. In addition, first principle simulation can be applied in SiO_2 brazing to prove a possible and clear reaction process. The thermodynamic statistics that simulation offers can fill the blank of thermodynamic statistics of some reaction products, such as enthalpy and Gibbs energy. The accurate reaction process in the brazing may be unveiled by further study.

3.3. Control of residual stress

As mentioned in introduction part, the CTE of SiO_2 materials is around 0.55 × 10^{-6} which is quite low compared with other materials. Since brazing is usually used to joining dissimilar materials, the mismatch of CTE between base materials must be considered. Even in self-joining of SiO_2, CTE mismatch between base material and brazing filling materials also need to be taken into consideration. In brazing process, when brazing filling materials start to melt and react with base materials as temperature rises, base materials expand without restraints. But as melting brazing filling materials solidify, the different shrinkages of each material will cause the large residual stress. This will largely impair the strength of joints. Even worse, once residual stress accumulates to a high level which leads to cracks or even directly breaks the joining, the brazing process will fail. Therefore, in silica brazing research, control of residual stress becomes one indispensable part. Up to now, the control of residual stress can be divided into two paths, including adding materials with good plasticity which can easily deform under brazing condition and adjusting CTE of brazing filling materials to fit CTE of SiO_2 or form a gradient transition area.

Starting with stress relief through deformation, brazing alloy can relieve part of brazing stress by deformation since most metals, especially Cu, have a good plasticity. Cu often functionalizes as soft material in AgCuTi to alleviate stress through deformation. Therefore, utilizing AgCuTi brazing alloy, SiO_2 (or SiO_{2f}/SiO_2) has been successfully joined with copper, Nb, TC4 and 30Cr3 high-tensile steel. Nevertheless, to further elevate joint strength, other measures should also be taken to alleviate residual stress.

Firstly, due to good plasticity of Cu, insert a Cu layer into brazing seam is one good strategy to alleviate residual stress. Designed AgCu/Cu/AgCuTi composite interlayer was applied to improve brazing of invar alloy and SiO_2-BN [92]. Fig. 10(a) shows that initial straight fringe of middle Cu layer became rough. The result demonstrated deformation of Cu layer under the effect of residual stress. As a result, the shear strength of joints increases 207%. The deformation ability of each component can be simply assessed by nanoindentation tests, shown in Fig. 10(d). plastic factor η can be defined as fraction of area of real displacement curve (such as OPB) on area surrounded by first half displacement curve and perpendicular line from apex (such as OPC). For example, η_{Fe2Ti}= A_{OPB}/A_{OPC}, A_{OPC} and A_{OPB} refer to each corresponding area. The larger plastic factor is, the better plasticity is [68]. This gives a simple calculation principle to reasonably choose formed phase so as to relieve thermal stress generated in brazing cooling process. The addition of foam metals is the further method to relieve residual stress [94]. Foam metals, as newly emerged materials in recent years, own unique 3D frame structure, shown in Fig. 10(b). This specific structure makes designed materials easy to realize deformation under the effect of stress while remaining origin metals' properties. Comparation of brazing samples with different interlayers (none, Cu foil and Cu foam) showed that the joint strength with Cu foam increased several times than that with/without Cu foil, proving that Cu foam interlayer can effectively reduce residual stress through the deformation of its framework [89]. Moreover, interfacial microstructure of joints with Cu foam in Fig. 10(c) exhibits that phase distributes evenly in brazing seam. This distribution of phase can avoid the aggregation of brittle phase. However, foam metals are also facing the disintegration problems due to the dissolving and reaction. To keep the intact structure of foam metals, coating CNT on foam metal surface is one viable approach [81]. Inert CNT effectively limits the reaction between Ni and Ti. The approaches of adding metal layer are listed in Table 5.

On the other hand, the adjustment of brazing filling materials' CTE is a more common way than deformation to control residual stress. The brazing filling material is usually not a single element or metal but the combination of several elements. This leaves spaces for researchers to look for brazing filling materials with suitable CTE by altering the brazing filling materials components. Because carbon-based materials own low or negative CTE and extremely stable, they become perfect additives in SiO_2 brazing. As shown in Fig. 11(a), application of graphene or carbon nanotube (CNT) in this aspect has been widely studied by researchers. For example, CNT reinforced TiNi brazing alloy was applied to join Nb and SiO_2-BN. Since CNT owns a negative CTE (-5.86 × 10^{-9} /K), it effectively constrained the expansion of TiNi matrix (15.4 × 10^{-6} /K). Besides, there is another residual stress control mechanism of CNT. In the brazing seam of TC4 and SiO_2-BN, added CNTs acts as nucleation site to refine phase in brazing seam [85]. Interestingly, graphene can also reduce the seam CTE by increasing Cu(s, s) contents in the seam so as to reduce residual stress through defor-

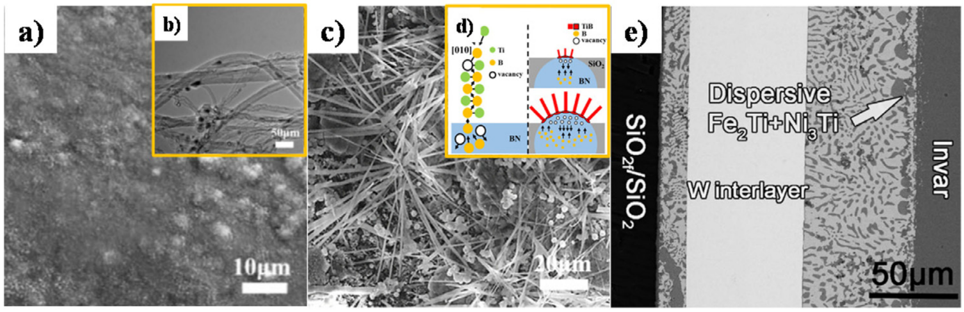

Fig. 11. Adjustment of CTE. (a) (b) CNT addition [85] (c) in-situ formation of TiB (d) schematic mechanism of TiB formation [95] (e) W interlayer microstructure after baring [72].

Table 6
Shear strength of SiO_2 brazing joints.

Brazing filling materials	Base materials	Improvements	Joint shear strength/ MPa	Joint size/ mm^2	Refs.
AgCuTi alloy	SiO_{2f}/SiO_2 / SiO_{2f}/SiO_2	Carbothermal reduction reaction	19	5 × 5	[76]
AgCuTi alloy	SiO_{2f}/SiO_2 / SiO_{2f}/SiO_2	VA-CNT	17	5 × 5	[78]
AgCuTi alloy	SiO_{2f}/SiO_2 / Al_2O_3	-	38.6	4 × 5	[98]
AgCuTi alloy	SiO_{2f}/SiO_2 / Nb	Surface etching	52.9	5 × 5	[82]
AgCuTi alloy	SiO_{2f}/SiO_2 / Nb	Plasma treatment	60.5	5 × 5	[75]
AgCuTi alloy	SiO_{2f}/SiO_2 / invar	-	26	5 × 5	[91]
AgCuTi alloy	SiO_{2f}/SiO_2 / invar	Graphene modified	26	5 × 5	[10]
AgCuTi alloy	SiO_2-BN / invar	-	32	3 × 3	[19]
AgCuTi alloy	Macor (46% SiO_2) / Ti	-	68	φ13	[99]
AgCuTi alloy	SiO_2 / TC4	-	27	5 × 5	[87]
AgCuTi alloy	SiO_2 / Cu	-	22	5 × 5	[100]
AgCuTi alloy	SiO_2 / 30Cr3 steel	-	37	5 × 5	[9]
AgCu-4.5 wt.%Ti/W/AgCu-1 wt.%Ti	SiO_{2f}/SiO_2 / invar	W interlayer	33	5 × 5	[72]
Ag-27.5Cu-2.5Ti powder with (h-BN)	SiO_2-BN / Ti	BN particles	31.4	3 × 3	[96]
AgCuInTi alloy	SiO_{2f}/SiO_2 / Nb	-	30.9	φ5	[101]
AgCuInTi alloy	SiO_{2f}/SiO_2 / TC4	-	19.6	-	[11]
AgCu alloy	SiO_2-BN / SiO_2-BN	electron-beam evaporated Ti	39.2	4 × 4	[102]
AgCu alloy	SiO_{2f}/SiO_2 / invar	Metallized by Ni	29	Less than 10 × 10	[103]
AgCu/Ni	SiO_2(74%) / TC4	-	110	-	[90]
AgCu/Cu/AgCuTi	SiO_2-BN / invar	Cu interlayer	43	5 × 5	[92]
AgCuNi + nano Al_2O_3 powder	SiO_2(76%) / TC4	-	40	5 × 5	[97]
Cu-23Ti powder	SiO_{2f}/SiO_2 / invar	graphene-modified	15	4 × 4	[84]
TiNi	SiO_2-BN / Nb	CNT reinforced	84	-	[80]
TiZrNiCu	SiO_2 / TC4	-	23	5 × 5	[104]
TiZrNiCu	SiO_2-BN / TC4	In-situ synthesized CNTs	35.3	5 × 5	[85]
TiZrNiCu	SiO_2-BN / TC4	Surface etching	29.7	5 × 5	[95]
TiZrNiCu/Ni foam	SiO_2-BN / TC4	CNTs-reinforced (Ni foam)	50	5 × 5	[81]
$Sn_{3.5}Ag_4Ti$(Ce,Ga) alloy	SiO_{2f}/SiO_2 / SiO_{2f}/SiO_2	-	17.91	5 × 5	[105]

mation as mentioned before [84]. On the other hand, adding or in-situ formation of new materials can work as gradient buffer to alleviate the residual stress, as shown in Fig. 11(c). Due to large CTE difference between silica and other materials, there is plenty space for researchers to find a suitable additive without introducing harmful effects. According to the different components of base materials, additional Al_2O_3 and TiB are beneficial to buffering the CTE mismatch, for instance [95–97]. The approximate CTE of brazing seam can be assessed by average CTE value of all components, as formula [6],

$$\alpha_{bs} = \alpha_m\left(1 - \sum V_i\right) + \sum \alpha_i V_i \quad (6)$$

α_{bs} is the CTE of whole brazing seam, α_m is original CTE of chosen brazing filling metal, V_i is volume content of additive i and α_i is CTE of additive i. This could give researchers a semiquantitative way to predict the brazing seam CTE. According to this formula, Yang mentioned addition of 3wt% BN could sharply reduce 21% of the CTE of brazing seam [96]. This proves the validity of additive in adjusting CTE. However, it is worth mentioning that the size of additive particle has an influence on the extent of CTE adjustment. Nanoparticles usually have a better improvement than macroparticle.

At last, inserting one interlayer which can remain intact is also viable to solve CTE mismatch in SiO_2 brazing. For example, W interlayer which can hardly react with AgCuTi brazing filling materials achieves CTE gradient changing. As can be seen from Fig. 11(e), W interlayer remains unchanged after brazing. And W interlayer with a medium CTE shows a large increasement in joint strengths.

Generally speaking, deformation and CTE adjustment do not singly work in the brazing process. In each brazing experiment, these two mechanisms synergistically act. It is hard to explain the accurate effects of each mechanism. Therefore, most research in this area still remains at discussing the external performance. Currently, the assessment of residual stress is the corresponding mechanical performance of the joints. It is highly recommended that some characterization methods which can directly show the level of residual stress, such as Raman and XRD, are added to explore the residual stress. Especially, in-situ methods are extremely useful in this aspect. If the accurate assessment of residual stress was found, the further optimization of brazing is imminent.

3.4. Mechanical performance of SiO_2 joint

The ultimate goal of brazing is to achieve a robust joining. And the most direct assessment is the mechanical performance. In addition, mechanical performance is also the reflection of interfacial microstructure and thermal residual stress. Therefore, the reported mechanical performance of brazing joints is listed in Table 6.

According to Table 6, it is clear that AgCuTi is still the most widely used brazing filling metal in brazing SiO_2 based materials. Direct usage of AgCuTi can achieve the bonding of heterogeneous materials. The shear strength of the joints is highly related to the difference of the CTE of base materials. The difficulty of joining increases with the increasement of the content of SiO_2 in base materials. Those additives in SiO_2 based materials reduce the CTE difference and improve the interfacial structure at the braze seam. SiO_{2f}/SiO_2 composites are hard to gain satisfying joint strength, so surface modifications are applied in these materials. The difference of the highest joint strength in each paper is related to the condition of SiO_{2f}/SiO_2 composites. Even with the same structure, the shear strength of the SiO_{2f}/SiO_2 composites itself varies from 40 MPa to 80 MPa at room temperature.

In addition, the joint size of reported sample is much lower than actual application of SiO_{2f}/SiO_2 composites. The small-size joints allow the mechanical tests but cannot accurately reflect the joining of the large-size samples. The accumulation of thermal stress in larger size would ruin the strength of the joints. Those residual stress control methods should be further investigated to gain a viable range of joining area. However, compared with other methods, the joint strength of brazing sample still owns obvious advantages.

The different brazing filling materials show a similar joint strength in joining the same base materials. But it is worth mentioning that different brazing filling materials have different usage temperatures. AgCuTi is recommended to be used at the temperature lower than 400°C. The limitation of TiZrNiCu is usually 600°C. And TiNi can withstand the highest temperature (around 850°C). The choice of brazing filling materials should take the application, interfacial microstructure and residual stress into consideration.

4. Conclusion and perspective

SiO_2 is one necessary industrial material which needs to be joined to itself or other materials. In all silica joining technologies, laser welding, anodic bonding and wafer bonding are emphasized in detail since each of them has unique advantage in specific area. Laser welding has obvious advantages in transparent SiO_2 glass joining by non-linear adsorption. Anodic bonding can achieve a relatively strong joining at low temperatures. And wafer bonding is extremely helpful to achieve joining of functional products with small size. According to vast investigations, brazing is the hottest topic in SiO_2 based ceramics joining since it can easily achieve joining of samples with complex shape and various sizes. Further, it is the only reported method that can achieve the joining of SiO_{2f}/SiO_2 composites. The brazing of SiO_2 based ceramics has been elaborated in its wetting, residual stress control, microstructure and mechanical performance. It owns obvious advantages in joint strength, flexibility and general applicability. Based on aforementioned research, some expectations have been proposed:

(1) Laser welding of SiO_2 based materials may be further extended to non-transparent ceramics since Penilla at al. has successfully applied laser welding in non-transparent ceramics materials by adjustments of non-linear absorption and linear absorption [106]. This break-through technology makes laser welding become a potential versatile SiO_2 joining method.

(2) Residual stress control needs some deeper exploration. Due to lack of viable test to assess residual stress, most research just mentioned the chosen way of relieving residual stress without a quantitative or semi-quantitative calculation. Applying more materialogy knowledge and simulation maybe helpful to investigate the deeper mechanism in this part. In addition, new methods or more advanced in-situ characterization technologies, such as in-situ Raman, synchrotron radiation and neutron radiation, are urgently needed to quantitively analyze residual stress.

(3) Non-metal brazing filling materials need to be developed in SiO_2-based ceramics joining. Glass brazing filling materials which have been applied in many ceramics brazing [107–109] are excellent candidates in SiO_2 based materials brazing. Most components in glasses are oxides, which means there is no difference in chemical bonding type between glass and SiO_2 based ceramics. A good wettability can be promised. In addition, glass matches many properties of silica, such as insulation, wave permeability and high temperature resistance. The low-self joining strength of SiO_{2f}/SiO_2 composites can be solved by this new type of brazing filling materials. If these new brazing filling materials were developed and utilized, brazing will be potential repairing technology of SiO_2 based materials and no more limited in joining aspects.

(4) Further application of first principle simulation in joining of SiO_2 materials should be useful. Though interfacial products have been clear in joining of SiO_2 materials, the accurate reaction paths in this process remain vague. Theoretical analysis needs to be reinforced to further elucidate joining mechanism, especially at atomic level. First principle simulation could offer atomic bonding information and verify the possible reaction process. DFT calculation has been applied in joining of other materials to understand the interfacial reactions [110–112]. This technology may be a new guidance on joining of SiO_2 materials. A successful simulation system is highly possible to predict joining results.

(5) The after-joining tests are recommended, especially for functional components. Except for strength assessments, more functional tests are required to evaluate the effects of the joints. For example, the dielectric properties of SiO_{2f}/SiO_2 composites needs to be remained since these composites are radome materials. But the applied brazing filler are mostly metals which will definitely damage dielectric properties of SiO_{2f}/SiO_2 composites. Therefore, after-joining tests needs to be supplemented.

Conflicts of interest

There are no conflicts to declare.

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